

SHIVAM CHAND KAUSHIK

AI/ML Engineer | Semiconductor Domain Expert

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Professional Summary

AI/ML engineer with 9 years of semiconductor and systems software experience spanning advanced packaging, IC inspection, and factory automation at **Kulicke & Soffa** and **KLA**. Currently pursuing **M.Tech in Artificial Intelligence from IIT Jodhpur**. Building production-grade AI systems—multi-agent architectures, vision-language models, and knowledge graph platforms—specifically for semiconductor manufacturing, inspection, and EDA workflows. One of a rare cohort who combines deep fab-floor domain knowledge (SECS/GEM, wire bonding, pick-and-place, machine vision) with modern AI/ML engineering (LangGraph, PyTorch, LLaVA, RAG, multi-agent orchestration). Proven ability to ship production C++ software across three countries, lead engineering teams, and deliver systems deployed to military and semiconductor customers.

Core Competencies

AI/ML & Deep Learning	Semiconductor Domain	Software & Infrastructure
PyTorch, TensorFlow, torchvision LangGraph, LangChain, RAG LLaVA, Qwen-VL, DeepSeek-Coder XGBoost, scikit-learn, SHAP Computer Vision, CNNs, ViT Multi-Agent Orchestration	Wafer Defect Inspection (KLA) Adv. Packaging & Wire Bonding (K&S) SECS/GEM Protocol (Intel integration) Pick & Place (iFlex, Luminex, Picalux) SPC, Yield Analysis, Pred. Maintenance Smart Factory / Industry 4.0	C++ (9 years production) Python FastAPI React, Streamlit Git/CI-CD

Work Experience

Software Engineer — Semiconductor Assembly & Smart Factory AI

Kulicke & Soffa (K&S), Netherlands & Singapore

Sept 2020 — Present

via ASM Technologies | K&S is a global leader in semiconductor assembly equipment for advanced packaging.

- Developing and optimizing the C++ software stack for RapidPro and ATPP wire bonding machines, engineering real-time machine control systems that process high-frequency sensor data and manage process parameter optimization.
- Architected product decoupling in the Machine Configuration Tool (MCT), enabling multi-platform support across 6 machine variants (iFlex-T2/T4/H1/CX, Luminex, Picalux).
- Integrated **MQTT protocol** into MCT for the Smart Factory initiative, enabling real-time machine-to-cloud data streaming for production analytics.
- Led code quality modernization: compiler warning upgrades (W3 to W4), static analysis (PVS-Studio), IWYU implementation, and Boost-to-STL migration.
- Applied AI to K&S semiconductor work: built an **LLM-powered SECS/GEM Protocol Assistant (RAG)** leveraging deep K&S equipment protocol knowledge, and an **ML-based Wire Bond Quality Prediction** system with Bayesian process optimization modeled on RapidPro/ATPP machine parameters—reducing predicted defect rate by 40%.

Software Engineer — IC Inspection & Machine Vision

Quest Global Technologies (client: KLA/ICOS)

July 2018 — Sept 2020

Bangalore | KLA is the world's leading semiconductor process control and inspection equipment company.

- Built a Python-based hardware simulator for the ICOS T800 Gantry System—a TCP/IP server modeling machine behavior for offline testing and digital-twin development.

- Developed data logging and analytics pipelines for production metrics: throughput tracking, taper parameter monitoring, and per-head performance analysis.
- Implemented **SECS/GEM interface for Intel**—the semiconductor industry standard protocol for equipment-to-host communication in automated fabs.
- Engineered granular process control: independent per-head yield-loss diagnostics for YZ1/YZ2/YZ4 pick & place heads.

Software Engineer — Telecom & Embedded Systems

Quest Global Technologies (client: Reliance Jio)

2018

Bangalore

- Developed LibWebSocket server for IoT smart speaker integration with ONT router—embedded systems communication architecture for Jio's IoT ecosystem.

Software Engineer — Aerospace & Defense Systems

AxisCades Aerospace & Technology

May 2016 — June 2018

Bangalore

- Developed real-time 2D signal visualization for a Bird Detection Radar system using C++/Qt.
- **Led a team of 5 engineers** and commissioned ACRT across **37 units of the Indian Army and Indian Air Force**.

AI/ML Projects

1. Multi-Modal Vision-Language AI for Semiconductor Inspection & Autonomous FA Report Generation [GitHub](#) | [HuggingFace Dataset](#)

- Built a four-agent LangGraph pipeline (Defect Descriptor, Root Cause Analyzer, Severity Classifier, Recipe Advisor) that fuses DINOv2 visual embeddings, SECS/GEM-format equipment telemetry, and a Qdrant historical defect retrieval database into a unified LLaVA-1.6 context—generating structured Failure Analysis reports in 48 seconds, replacing a 2–4 hour manual process.
- Trained a DINOv2 + MLP defect classifier on SemiFA-930, a self-constructed 930-image dataset spanning 9 semiconductor defect classes (SEM, optical, wafer map); achieved 92.1% accuracy / 0.917 Macro F1 with only 214K trainable parameters vs. 23.5M for ResNet-50 (82.9%).
- Publicly released SemiFA-930 dataset on HuggingFace; paper prepared for arXiv submission (cs.CV).
- **Stack:** LLaVA-1.6, DINOv2, QLoRA/PEFT, LangGraph, SECS/GEM, Qdrant, FastAPI, ReportLab, Streamlit

2. Wafer Defect Detection & Classification (WM-811K) [GitHub](#)

- Built an end-to-end deep learning pipeline for wafer defect pattern classification on the WM-811K dataset (811K+ wafer maps, 9 defect classes) using ResNet18 and EfficientNet-B0 architectures, with a ConvAutoEncoder for anomaly detection on unseen defect patterns. Achieved 96.2% overall test accuracy and 90.5% defect-class accuracy on a severely class-imbalanced dataset (103,201 normal vs 104 Near-Full samples); resolved class collapse via WeightedRandomSampler and diagnosed 6 critical training bugs across normalization, BatchNorm domain shift, and loss double-suppression.
- Deployed as a FastAPI inference server with `/classify` and `/anomaly-detect` endpoints; live end-to-end API testing confirmed 91–94% confidence across 7 of 8 defect classes on real WM-811K wafer maps. Anomaly threshold calibrated at the 95th percentile of normal-wafer reconstruction error.
- **Stack:** PyTorch, torchvision, scikit-learn, FastAPI

3. LLM-Powered SECS/GEM Protocol Assistant (RAG) (*Applied at Kulicke & Soffa*)

- Built a RAG-powered LLM assistant for the SECS/GEM semiconductor protocol, enabling natural language queries over SEMI standards, equipment specifications, and alarm codes. Fine-tuned retrieval on SECS/GEM documentation with FAISS vector store.
- Deployed as an interactive web application for equipment and process engineers in semiconductor fabs.
- **Stack:** HuggingFace Transformers, LangChain, FAISS, Streamlit

4. **Wire Bond Quality Prediction & Process Optimization** *(Applied at Kulicke & Soffa)*

- Developed ML-based wire bond quality prediction using process data (bond force, ultrasonic power, temperature, loop height) modeled on K&S RapidPro and ATPP machine parameters. Classified bond pass/fail with root cause analysis and applied Bayesian optimization to identify optimal bonding parameters, reducing predicted defect rate by 40%.
- **Stack:** scikit-learn, Optuna, matplotlib

5. **Agentic AI for HDL Code Analysis, Generation & Verification** *(Applied at Kulicke & Soffa)*

- Architected a multi-agent system for semiconductor HDL verification: four specialized agents (RTL Analyzer, Testbench Generator, Lint/DRC, Design Space Explorer) autonomously parse Verilog/SystemVerilog, generate UVM testbenches with self-correction loops, and optimize design parameters via reinforcement learning—targeting 60–70% of the chip development cycle spent on verification.
- Integrated open-source EDA tools (Icarus Verilog, Yosys, Verilator) as agent tools with RAG over IEEE Verilog standards and design pattern libraries. Supports fully air-gapped deployment with local LLMs for semiconductor IP protection.
- **Stack:** LangGraph, DeepSeek-Coder, pyverilog, Icarus Verilog, Yosys, Qdrant, Stable-Baselines3, Optuna, FastAPI, React, Docker

Education

Indian Institute of Technology (IIT), Jodhpur

June 2024 — May 2026

Master of Technology (M.Tech) — Artificial Intelligence

JSS Academy of Technical Education, Noida

June 2011 — May 2015

Bachelor of Technology (B.Tech) — Information Technology

Achievements

- **Bright Mind Award** — ASM Technologies Ltd. (2021), recognizing exceptional technical contribution in semiconductor software engineering
- **779th Rank** — International Mathematics Olympiad (IMO), demonstrating strong analytical and quantitative aptitude

Certifications (In Progress)

Certification	Provider
Fundamentals of Deep Learning	NVIDIA Deep Learning Institute (DLI)
AWS Certified Machine Learning Engineer — Associate	Amazon Web Services